Synthesis and polymerisation of new multifunctional urethane methacrylates

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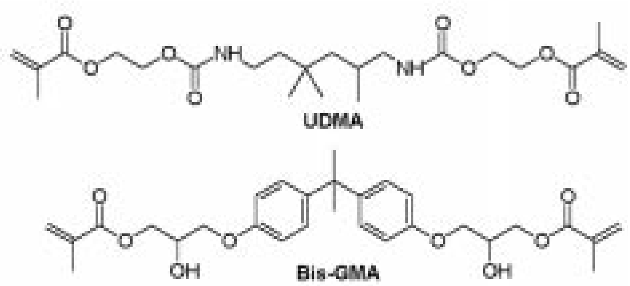
SUMMARY: New multifunctional urethane methacrylates were synthesised by addition of methacrylates containing a hydroxy group, such as 2-hydroxyethyl methacrylate, 2-hydroxypropyl methacrylate or glycerol dimethacrylate, to α,α,α′,α′-tetramethyl-m-xylylene diisocyanate. The structure of the new monomers was confirmed by IR, ^{1}H NMR and ^{13}C NMR spectroscopy. The photopolymerisation of the crosslinking monomers in bulk using camphorquinone as photoinitiator results in transparent, crosslinked polymers.

ZUSAMMENFASSUNG: Neue multifunktionelle Urethanmethacrylate wurden durch Addition von hydroxygruppenhaltigen Methacrylaten, wie 2-Hydroxyethylmethacrylat, 2-Hydroxypropylmethacrylat oder Glycerindimethacrylat, an α,α,α′,α′-Tetramethyl-m-xylylendiisocyanat hergestellt. Die Struktur der neuen Monomeren konnte durch IR-, ^{1}H NMR- und ^{13}C NMR-Spektroskopie bestätigt werden. Die Photopolymerisation der Vernetzermonomeren in Substanz führte mit Campherchinon als Photoinitiator zu transparenten Polymernetzwerken.

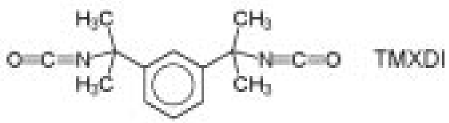
Introduction

UV-curable monomeric or oligomeric urethane acrylates or methacrylates are used as coating or resin component and in stereolithography[1,2]. Furthermore, urethane dimethacrylates are well established in dentistry as resin monomers for dental composites[3–5]. The most commonly used dental monomer of this type is 1,6-bis-(2-methacryloyloxyethoxycarbonylamino)-2,4,4-trimethylhexane (UDMA, Scheme 1), the reaction product of 2 mol 2-hydroxyethyl methacrylate (HEMA) with 1 mol 2,4,4-trimethylhexamethylene diisocyanate (TMDI). The photopolymerisation of UDMA alone results in more flexible materials. Therefore, UDMA is mostly used in combination with the less flexible 2,2-bis[4-(2-hydroxy-3-methacryloyloxypropyl)-phenyl]propane (Bis-GMA, Scheme 1)[6], which is a mixture of isomers. From a practical point of view, it would be an advantage to use a more rigid urethane dimethacrylate. Therefore, we tried to prepare multifunctional urethane

Scheme 1.



Scheme 2.



methacrylates which result in materials with mechanical properties comparable with those of Bis-GMA-based resins.

α,α,α′,α′-Tetramethyl-m-xylylene diisocyanate (TMXDI) (Scheme 2) is a commercially available diisocyanate produced in non-phosgene processes[7]. Because of its unsymmetrical structure, crystallinity common in corresponding urethanes is hindered. The diisocyanate would be expected to yield rather liquid urethanes. UV-curable oligomeric urethane acrylates were obtained by the reaction of α,ω-dihydroxypoly(caprolactone) or polyamine with TMXDI and HEMA[8,9].

Previously, we reported about the synthesis and polymerisation of crosslinking 2-vinylcyclopropanes[10,11]. This paper describes the synthesis and photopolymerisation of new multifunctional urethane methacrylates **1a**–**e** (Scheme 3) and composites based thereupon.

Experimental

Materials

HEMA, 2-hydroxypropyl methacrylate (HPMA), dichloromethane, glycidyl phenyl ether, ethyl acetate, hexane and tet-

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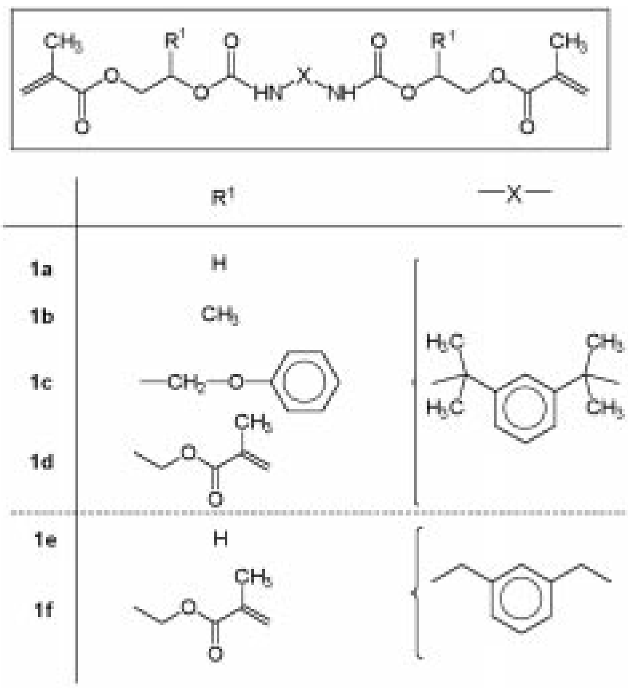
Scheme 3.



rahydrofuran (THF) were dried over molecular sieves. Glycerol dimethacrylate (GDMA, Röhm) was dried with anhydrous sodium sulfate. GDMA is an isomeric mixture of 37.6% of 1,2- and 60.8% of 1,3-diester. m-Xylylene diisocyanate (XDI, Takeda), and TMXDI (Cytec) were used without further purification. Benzyl α-hydroxymethylacrylate was prepared analogous to the literature[12]. 2-Hydroxy-3-phenoxypropyl methacrylate was prepared by adding methacrylic acid to glycidyl phenyl ether[13]. Dibenzoyl peroxide (DBPO, Fluka) was purified by recrystallisation from chloroform. 4-Dimethylaminopyridine (DMAP), dibutyltin dioctanoate (DBTDO, Acima), bisphenol A glycidyl dimethacrylate (Bis-GMA, Esschem), triethylene glycol dimethacrylate (TEGDMA, Esschem), aliphatic urethane dimethacrylate (UDMA, Ivoclar, Liechtenstein), DL-camphorquinone (CC, Rahn, Zürich, Switzerland), and N-(2-cyanoethyl)-N-methylaniline (CEMA, Lonza, Basel, Switzerland) were used without further purification.

The fillers are commercially available products: fumed silica (Degussa, Frankfurt, Germany), YbF_3 (Auer Remy, Hamburg, Germany), SiO_2—ZrO_2 mixed oxide (Tokoyama Soda), and barium silicate glass (Schott, Landshut, Germany). Before use, the silica, mixed oxide, and glass filler were modified with the silane coupling agent 3-methacryloyloxypropyltrimethoxysilane (Union Carbide). The fillers were silanised by mixing them with 1.0 wt.-% of water and 5.0 wt.-% of silane at room temperature over a period of two h, and drying the modified fillers at 50 °C for 4 days.

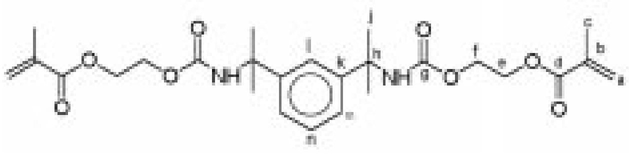
Synthesis of the urethane methacrylate monomers (**1**) *(general procedure)*

125 mmol diisocyanate is added to a solution of 251 mmol of hydroxyalkyl methacrylate and 50 mg of DBTDO at room temperature. The formed solution is stirred for 36 h at 70 °C under reflux. Once the isocyanate is completely consumed (IR-spectroscopy), the reaction mixture is diluted with 500 mL of anhydrous methylene dichloride, extracted 3 times with 100 mL of aqueous 1M NaOH, washed 3 times with water, and dried with anhydrous sodium sulfate. Then, the solvent is removed on a rotary evaporator under reduced pressure.

1a: Addition product of HEMA to TMXDI; yield 88% of a colourless liquid, η (23 °C) = 860 Pa · s, η (40 °C) = 40 Pa · s, n_D^{25} = 1,5120; it crystallises to a solid compound (m.p. 53 °C) after standing for about 3 days in the refrigerator.

IR (neat): 3361 (s), 2976 (s), 1714 (s), 1637 (m), 1504 (s), 1384 (s), 1174 (s), 1043 (m), 944 (m) cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 1.66 (s, 12 H, CH_3), 1.95 (s, 6 H, $=C-CH_3$), 4.24–4.40 (br, 8 H, OCH_2CH_2O), 5.29 (br, 2 H, NH), 5.77 and 6.13 (2 s, 4 H, $=CH_2$), 7.27–7.43 (m, 4 H, aromatic CH) ppm.

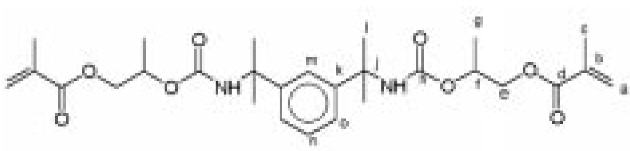


^{13}C NMR (100 MHz, $CDCl_3$): δ = 18.4 (j), 29.3 (c), 55.3 (h), 62.3 and 63.8 (e, f), 121.1 (l), 123.1 (m), 125.7 (a), 128.2 (n), 136.0 (b), 147.0 (k), 154.2 (g), 164.1 (d) ppm.

1b: Addition product of HPMA to TMXDI; yield 85% of a colourless, highly viscous liquid, η (23 °C) = 1670 Pa · s, n_D^{25} = 1,5018.

IR (neat): 3366 (m), 2980 (s), 1719 (s), 1637 (m), 1521 (s),1458 (s), 1296 (m), 1248 (s), 1172 (s), 1088 (m) cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 1.32 (d, *, $CHC\underline{H}_3$), 1.62 (s, 12 H, CH_3), 1.93 (s, 6 H, $=C-CH_3$), 4.09–4.29 (m, *, OCH_2), 5.00 (m, *, OCH), 5.14–5.17 (br, 2 H, NH), 5.69 and 6.22 (2 s, 4 H, $=CH_2$), 7.22–7.26 (m, 4 H, aromatic CH) ppm (Σ* = 12 H).



^{13}C NMR (100 MHz, $CDCl_3$): δ = 14.6 and 15.9 (g), 16.5 (l), 27.7 (c), 63.3, 66.0, 67.6 and 70.3 (e, f), 119.4 (m), 121.3 (o), 124.1 (a), 126.4 (n), 134.5 (b), 147.0 (k), 154.2 (h), 164.1 (d) ppm.

1c: Addition product of 2-hydroxy-3-phenoxypropyl methacrylate to TMXDI; yield 70% of a colourless, highly viscous liquid, η (23 °C) = 2150 Pa · s, n_D^{25} = 1,5413.

IR (neat): 3363 (s), 2976 (s), 1732 (s), 1600 (m), 1504 (s), 1164 (s), 1082 (m) cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 1.63 (s, 12 H, CH_3), 1.93 (s, 6 H, CH_3), 4.05–4.47 (m, 8 H, OCH_2), 5.16 (br, 2 H, NH), 5.28–5.29 (m, 2 H, OCH), 5.57 and 6.10 (2 s, 4 H, $=CH_2$), 6.89–7.40 (m, 14 H, aromatic CH) ppm.

^{13}C NMR (100 MHz, $CDCl_3$): δ = 18.3 (o), 29.5 (c), 55.5 (n), 62.7 (g), 66.4 (e), 68.7 (f), 114.6, 121.0, 123.2, 126.1, 126.4, 128.4, 129.5 and 135.8 (a, b, j, k, l, q, r, s), 146.8 (p), 154.0 (h), 158.4 (m), 166.9 (d).

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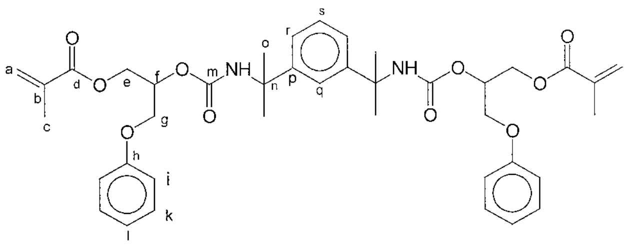
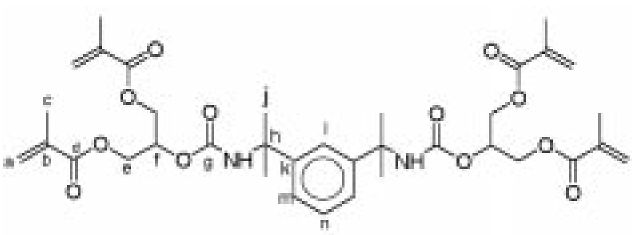


1d: Addition product of glycerol dimethacrylate to TMXDI; yield 84% of a colourless, highly viscous liquid, η(23 °C) = 241 Pa · s, n_D^{25} = 1,5069.

IR (neat): 3370 (s), 2973 (s), 1715 (s), 1638 (m), 1520 (s), 1455 (s), 1160 (s) cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 1.52 (s, 12H, CH_3), 1.91 (s, 12H, CH_3), 3.59–4.44 (m, 8H, CH_2O), 5.18–5.20 (m, 2H, CHO), 5.35 (br, 2H, NH), 5.57 and 6.12 (2 s, 8H, $=CH_2$), 7.25–7.41 (m, 4H, aromatic CH) ppm.

^{13}C NMR (100 MHz, $CDCl_3$): δ = 18.4 (j), 24.3 (c), 55.6 (h), 62.9 (e), 69.7 (f), 128.4 (a), 121.9, 126.4 and 128.4 (l, m, n), 136.1 (b), 147.3 (k), 155.4 (g), 166.6 (d) ppm.

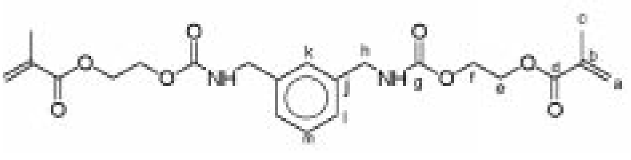


1e: Addition product of HEMA to XDI; yield 73% of a white solid (m.p. 95 °C).

IR (KBr): 3302 (m), 2938 (w), 1697 (s), 1636 (m), 1541 (s), 1460 (m), 1262 (s), 1174 (s) cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 1.88 (s, 6H, CH_3), 4.28–4.39 (br, 12H, CH_2N and CH_2O), 5.25 (br, 2H, NH), 5.59 and 6.14 (2 s, 4H, $=CH_2$), 7.19–7.46 (m, 4H, aromatic CH) ppm.

^{13}C NMR (100 MHz, $CDCl_3$): δ = 18.2 (c), 44.8 (h), 62.7 and 62.9 (e, f), 126.0 (a), 126.3 and 126.5 (k, l), 129.2 (b), 139.0 (j), 156.3 (g), 167.2 (d) ppm.

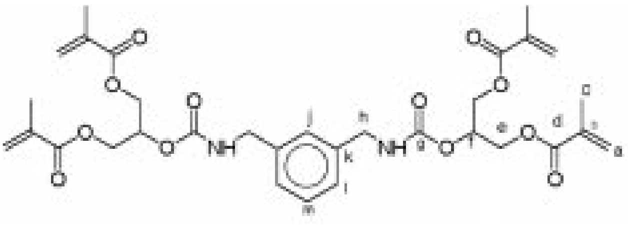


1f: Addition product of glycerol dimethacrylate to XDI; yield 70% of a colourless, highly viscous liquid, η(23 °C) = 575 Pa · s, n_D^{25} = 1,5127.

IR (neat): 3368 (m), 2958 (w), 1725 (s), 1636 (m), 1527 (s), 1163 (s) cm^{-1}.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 1.93 (s, 12H, CH_3), 4.22–4.40 (br, 12H, CH_2N and CH_2O), 5.29 (br, 2H, NH), 5.60 and 6.10 (2 s, 8H, $=CH_2$), 5.68–5.70 (m, 2H, OCH), 7.19–7.46 (m, 4H, aromatic CH) ppm.

^{13}C NMR (100 MHz, $CDCl_3$): δ = 16.8 (c), 44.1 (h), 61.5 (e), 68.5 (f), 125.0 (a), 125.0 and 125.1 (l, j), 127.5 (m), 134.4 (b), 142.1 (k), 154.6 (g), 165.4 (d) ppm.



Polymerisation

Photopolymerisation of monomer mixtures or composites was carried out using CC as the photoinitiator. Flexural strength specimens (2 × 2 × 25 mm) were obtained by irradiating the resins or the composites with a visible-light source (Spectramat, Ivoclar, Schaan, Liechtenstein). The mechanical properties were determined after immersing the specimens in water for 24 h at 37 °C.

Composite preparation

The experimental composites were mixed in a LPM 2 SP kneading machine (Linden, Marienheide).

Measurements

NMR spectra were recorded on a DPX-400 spectrometer (Bruker; ^{1}H: 400 MHz, ^{13}C: 100 MHz) using tetramethylsilane as the standard and $CDCl_3$ as the solvent. A FT-IR spectrometer 1600 (Perkin-Elmer) was used to record IR spectra. Differential scanning calorimetry (DSC) measurements were performed using a Perkin-Elmer DSC-7 thermal analyser with a photopolymerisation device. Scanning rates of 10 K min^{-1} were used. The flexural strength and flexural modulus were determined as usual using a Z 010 universal testing machine (Zwick, Ulm, Germany).

Results and discussion

The monomers **1a–f** were synthesised by adding hydroxyalkyl methacrylates to TMXDI (**1a–d**) or XDI (**1e**, **1f**) in the presence of DBTDO as catalyst (Scheme 4).

The monomers **1a** and **1e** are solids, while all of the other monomers are colourless, highly viscous liquids. The comparison of the melting points of **1a** and **1e** shows that the introduction of the four methyl groups leads to a significant decrease of the melting temperature by about 42 °C. In case of **1a**, the crystallisation at room temperature is very slow and can be accelerated by cooling. In contrast, the formation of solid **1a** can be suppressed by mixing **1a** with about 5–10 wt.-% of **1b**. As expected, the highest refractive index of the monomers was found in monomer **1c**, which contains three aromatic rings in the molecule.

The characterisation of the synthesised monomers was carried out by ^{1}H NMR, ^{13}C NMR and IR spectroscopy.

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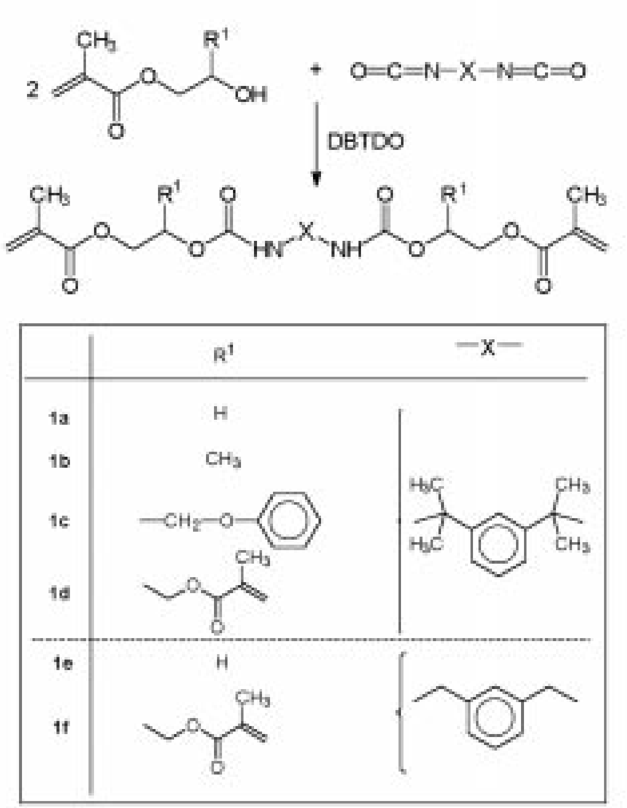
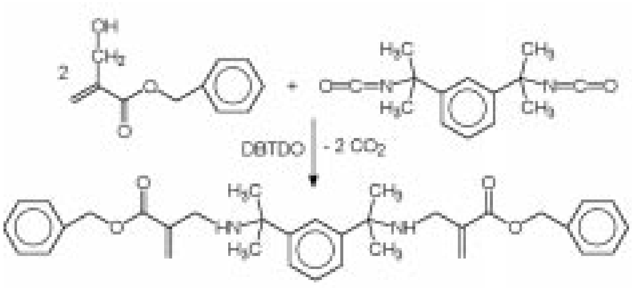
Scheme 4.



Scheme 5.



Tab. 1. Heat of the bulk polymerisation ΔH_P of the crosslinking monomers **1a–d** and **1f** with DBPO (2.5 wt.-%) determined by DSC.

Monomer	ΔH_P (kJ mol^{-1})	C═C conversion (%)[a]
1a	110.5	96
1b	108.9	94
1c	112,6	97
1d	158,6	69
1f	155.3	67
Bis-GMA	102.9	89
UDMA	115.1	99

[a] Calculated from the ΔH_P-values based on the standard heat of polymerisation of the methacrylate group of 57.8 kJ mol^{-1} [14].

Tab. 2. Mechanical properties of bulk polymerisates of mixtures of the crosslinking monomers **1a–f** (42.0 wt.-%) with UDMA (37.1 wt.-%) and TEGDMA (20.1 wt.-%); photoinitiator: CC/CEMA (0.8 wt.-%), irradition and polymerisation time: 6 min

Monomer	Flexural strength (MPa)		Flexural modulus (MPa)	
	dry[a]	wet[b]	dry[a]	wet[b]
1a	90	104	1910	2500
1b	89	84	2200	2400
1c	86	94	2040	2420
1d	86	85	2170	2510
1e	96	77	2320	2090
1f	72	58	1850	2060
Bis-GMA[c]	116	69	2550	2060

[a] Measurements of dry samples.
[b] Measurements of samples after storage for 6 days in water (37 °C) and for 1 day in boiling water.
[c] Content: 42.0 wt.-%.

The spectral data are in agreement with the expected structure. For example, the methacrylic groups are supported by the presence of two singlets assignable to at δ = 5.6–5.7 ppm and 6.1–6.2 ppm, or the broad signals of the urethane NH-protons at δ = 5.1–5.3 ppm in the ^{1}H NMR spectrum of the monomers **1a–f**.

Analogous to the synthesis of **1c**, we also tried the addition of benzyl α-hydroxymethylacrylate to TMXDI. During the reaction, the formation of carbon dioxide was observed. The ^{1}H NMR spectrum of the reaction product shows two unexpected signals: a broad peak at 1.67 ppm assignable to NH of a secondary amine and a singlet at 3,16 ppm assignable to the protons of a CH_2N-group. Furthermore, the product showed a significantly lower viscosity of 11.3 Pa · s (23 °C) than the other urethane dimethacrylates. Based on these results it can be assumed that a bifunctional amine methacrylate was formed (Scheme 5) by the elimination of carbon dioxide from the intermediary formed adduct of benzyl α-hydroxymethylacrylate and TMXDI (Scheme 5).

In order to determine the radical polymerisation behaviour of the new multifunctional methacrylates, the degree of double bond conversion was estimated by DSC. For the dimethacrylates **1a–c** a relatively high degree of double bond conversion was found (Tab. 1). In case of the tetrafunctional methacrylates, the accessibility of the methacrylate groups is confined as a result of the higher density of the formed network. Therefore the degree of double bond conversion of monomers **1d** and **1f** is significantly lower than that of the dimethacrylates **1a–c**. Because the synthesised monomers are both liquids (**1b**, **1c**, **1f**) and solids (**1a**, **1e**), the photopolymerisation of the monomers was carried out in mixtures of them with UDMA and TEGDMA as the diluent monomers, whereby a mixture of CC and CEMA was used as the photoinitiator. The results show (Tab. 2) that the flexural strength

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Tab. 3. Mechanical properties of composites[a] based on mixtures of the monomers **1a–f** (42.0 wt.-%) with UDMA (37.1 wt.-%) and TEGDMA (20.1 wt.-%); photoinitiator: CC/CEMA (0.8 wt.-%); irradiation and polymerisation time: 6 min

Monomer	Flexural strength (MPa)		Flexural modulus (MPa)	
	dry[b]	wet[c]	dry[b]	wet[c]
1a	141	143	13140	12920
1b	120	129	13620	11520
1d	127	107	11190	11060
1f	114	111	10860	10230
Bis-GMA	116	123	12130	10560

[a] Composition of the composites: 52.4 wt.-% of milled barium silicate glass, 12.8 wt.-% of YbF_3, 13.7 wt.-% of SiO_2-ZrO_2 mixed oxide, 1.0 wt.-% of silica, 20.1 wt.-% of the above mentioned monomer mixture.
[b] Measurements of dry samples.
[c] Measurements of samples after storage in water (37 °C) for 6 days and in boiling water for 1 day.

and flexural modulus of the photocured materials based on the monomers **1a–d** are less influenced by storage or boiling in water than their mixes with Bis-GMA. From the literature, it is known that the hydroxy groups of Bis-GMA are responsible for the water uptake and the corresponding decrease of the mechanical properties[6]. During the photopolymerisation of the monomer mixtures a similar volume shrinkage of about 7.4–7.8 vol.-% can be observed, which is not surprising given the similar molecular weight of the crosslinkers. From the monomers **1a**, **1b**, **1d** and **1f**, as well as Bis-GMA, particle filled composites were prepared and photocured. The results (Tab. 3) confirm that the synthesised crosslinkers, in particular monomer **1a**, can be used to substitute Bis-GMA in composites.

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